IntelliTrack - Capstone Submission Tracking System

A modern, AI-powered capstone project submission tracking system with analytics, group management, and smart reminders.

🚀 Features

- ✨ **AI-Powered Analytics** - Gemini AI integration for smart insights
- 👥 **Group Management** - Create, edit, and delete project groups
- 📋 **Deliverable Tracking** - Monitor submission statuses
- 🔐 **Role-based Access** - Admin, Coordinator, Adviser, and Student roles
- 🔄 **Google OAuth2** - Easy login with Google
- 📱 **Responsive UI** - Works on all devices
- 🛡️ **Production Ready** - Secure, scalable, and maintainable

🛠️ Tech Stack (Complete Versions)

Backend

Component	Version
Java	21
Spring Boot	3.2.0
Spring Data JPA	3.2.0
Spring Security	3.2.0
Spring OAuth2 Client	3.2.0
Spring Validation	3.2.0
Hibernate	6.4.0
MySQL Connector	8.4.0
PostgreSQL Driver	42.7.6
Maven	3.8+

Maven Compiler Plugin	3.11.0
Lombok	Latest
JJWT (JWT)	0.11.5
Jackson Datatype JSR310	Latest
AWS Java SDK S3	1.12.529
Caffeine Cache	Latest
Bucket4j (Rate Limiting)	7.6.0
Apache POI (Excel)	5.2.5
Dotenv Java	3.0.0

Frontend

Component	Version
React	18.2.0
React DOM	18.2.0
React Router DOM	6.8.0
Axios	1.16.0
Lucide React	1.14.0
Date-fns	4.1.0
Recharts	2.15.4
React Scripts	5.0.1
Node.js	18+
npm	9+

AI Integration

- Google Gemini 2.5 Flash

🎯 Quick Start

Prerequisites

- Java 21+
- Node.js 18+
- MySQL 8.0+ or PostgreSQL
- Maven 3.8+
- Git

1. Clone the Repository

```
git clone https://github.com/yourusername/intellitrack.git
cd intellitrack
```

2. Backend Setup

Step 1: Configure Environment

```
cd Backend
cp .env.example .env
```

Edit `.env` with your credentials:

```
# Required
GOOGLE_CLIENT_ID=your-google-client-id
GOOGLE_CLIENT_SECRET=your-google-client-secret
JWT_SECRET=your-secure-jwt-secret
AI_API_KEY=your-gemini-api-key

# Database
DB_HOST=localhost
DB_PORT=3306
DB_NAME=intellitrack
DB_USERNAME=root
DB_PASSWORD=yourpassword
```

```
# Server
SERVER_PORT=8080
```

Step 2: Create Database

```
CREATE DATABASE intellitrack CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci;
```

Step 3: Build & Run

```
mvn clean package -DskipTests
java -jar target/backend-0.0.1-SNAPSHOT.jar
```

Backend starts on http://localhost:8080

3. Frontend Setup

Step 1: Configure Environment

```
cd ../Frontend
echo "REACT_APP_API_URL=http://localhost:8080" > .env
```

Step 2: Install & Run

```
npm install
npm start
```

Frontend starts on http://localhost:3000

👤 Sample/Dummy Credentials

Role	Email	Password
Administrator	admin@intellitrack.com	IntelliTrack@2026!
Coordinator	coordinator@intellitrack.com	Coord@2026!
Adviser	adviser@intellitrack.com	Adviser@2026!

Student	student@intellitrack.com	`Student@2026!`

Docker Deployment

Run with Docker Compose

```
docker-compose up -d
```

This will start:

- MySQL database
- Backend API
- Frontend (Nginx)

Access:

- Frontend: http://localhost
- Backend: http://localhost:8080

Project Structure

```
intellitrack/
├── Backend/
│   ├── src/main/java/com/intellitrack/
│   │   ├── config/
│   │   ├── controller/
│   │   ├── dto/
│   │   ├── entity/
│   │   ├── repository/
│   │   └── service/
│   └── pom.xml
├── Frontend/
│   ├── src/
│   │   ├── components/
│   │   ├── pages/
│   │   ├── services/
│   │   └── App.js
│   └── package.json
└── README.md
```

🔧 Configuration

See detailed configuration in:

- Backend README
- Application Properties

🌓 Security

- JWT Authentication
- Google OAuth2 Login
- Role-based Access Control
- Input Validation
- SQL Injection Protection
- XSS Protection

🤝 Contributing

1. Fork the repository
2. Create your feature branch (`git checkout -b feature/AmazingFeature`)
3. Commit your changes (`git commit -m 'Add some AmazingFeature'`)
4. Push to the branch (`git push origin feature/AmazingFeature`)
5. Open a Pull Request

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📞 Support

For support, email jlo07391@gmail.com or open an issue.

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